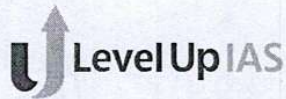


Q.3c (15 marks): How do people adapt to environmental extremes and other circumstances?



(Don't Write anything in this Area)

3.c) How do people adapt to environmental extremes and other circumstances? (15 marks)

Human are the most widespread species across the planet only because of their ability to adapt to various environmental conditions.

Ways people adapt to environmental extremes

1) Acclimatisation

It refers to temporary change that happen in the body's physiology that helps human to quickly respond to changed environment in short period of time.

It is reversible in nature as it doesn't involve any change in our genetics.

Ex) Acclimatisation to colder environment
 includes - higher rate of shivering
 - vasoconstriction at peripheral
blood vessels

2) Ontological acclimatisation

It refers to the ability of an individual to undergo such acclimatisation changes & not the change per say.

Exo Some individuals successful acclimatise to higher altitudes while others suffer from 'altitudinal sickness'

3) Adaptation

It refers to permanent changes that happen within human body that provides individual with survival advantage.

It is a long term change that is inherited across generations

Exo Tibetan population carry 'cry2' gene that helps them adapt well to Tibetan highlands.

Adaptation to high altitude - includes higher lung capacity, increased rate of haemoglobin production in blood, etc

4) Cultural adaptation

It refers to the changes one makes in his behaviours, norms or organisation in order to adapt to these stresses.

For ex British Columbia tribes make 'igloos' as their homes & use thick animal fur as their clothing to protect themselves against cold stress.

Malinowski has here called culture as a mean or tool to adjust to one's environment & its function of culture to help human survive better.

Therefore, both cultural & anatomic & behavioural aspects help human to adapt to different stresses.